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1 from sklearn.neural_network import MLPRegressor # подключаем модуль
2 from sklearn.datasets import make_regression # функция для генерации
3 from sklearn.model_selection import train_test_split # разделение на
4 X, y=make_regression(n_samples=2000, random_state=1) # создаем данные
5 X_train, X_test, y_train, y_test=train_test_split(X, y, #
6                                                    random_state=1) #
7 regr=MLPRegressor(random_state=1,
8                   hidden_layer_sizes=(15,15,), # количество нейронов
9                   #activation='tanh',
10                  max_iter=5000 # число эпох обучения
11                  ).fit(X_train, y_train) # создаем и обучаем регрессор
12 pred=regr.predict(X_test) # проверяем на тесте
13 pred_train=regr.predict(X_train)
14 regr.score(X_test, y_test) # смотрим на качество решения

```

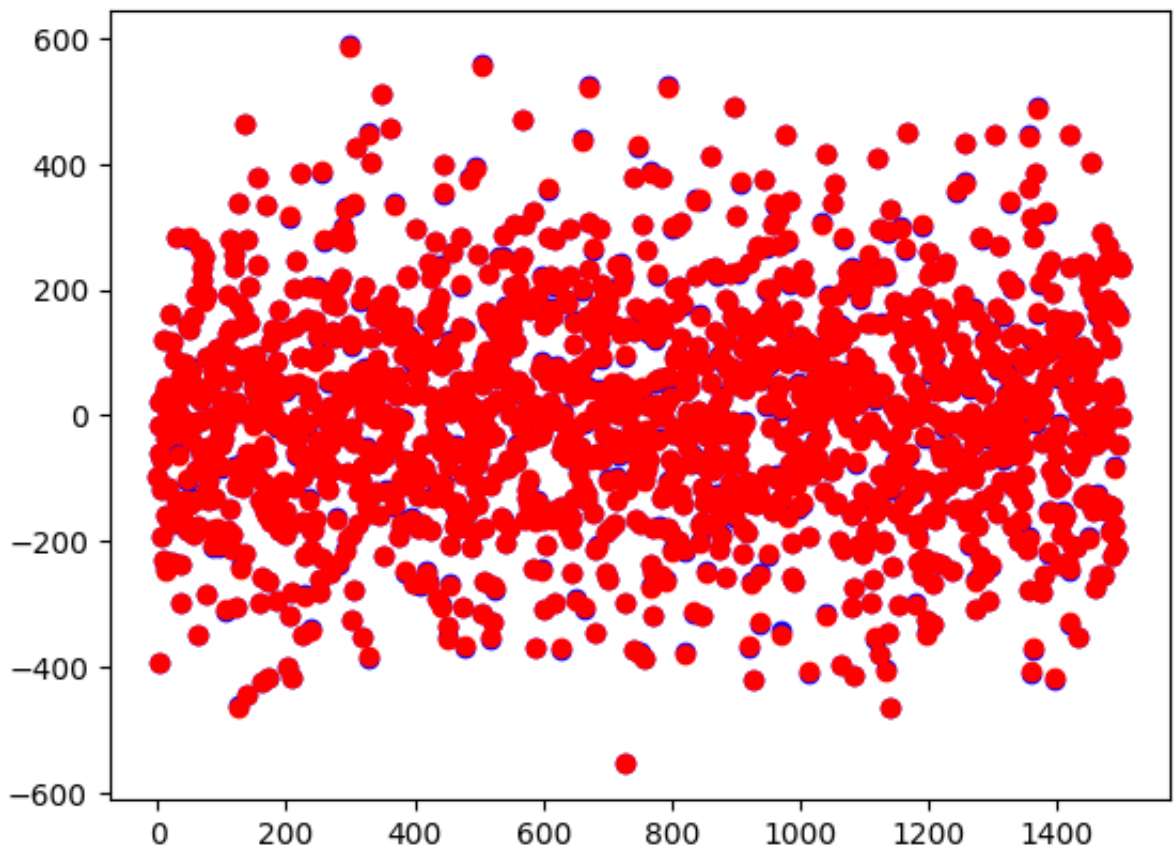
0.9934580546670959

```

1 # проверка обучения
2 from matplotlib import pyplot as plt
3 plt.scatter(range(1500),y_train,c='blue')
4 plt.scatter(range(1500),pred_train,c='red')

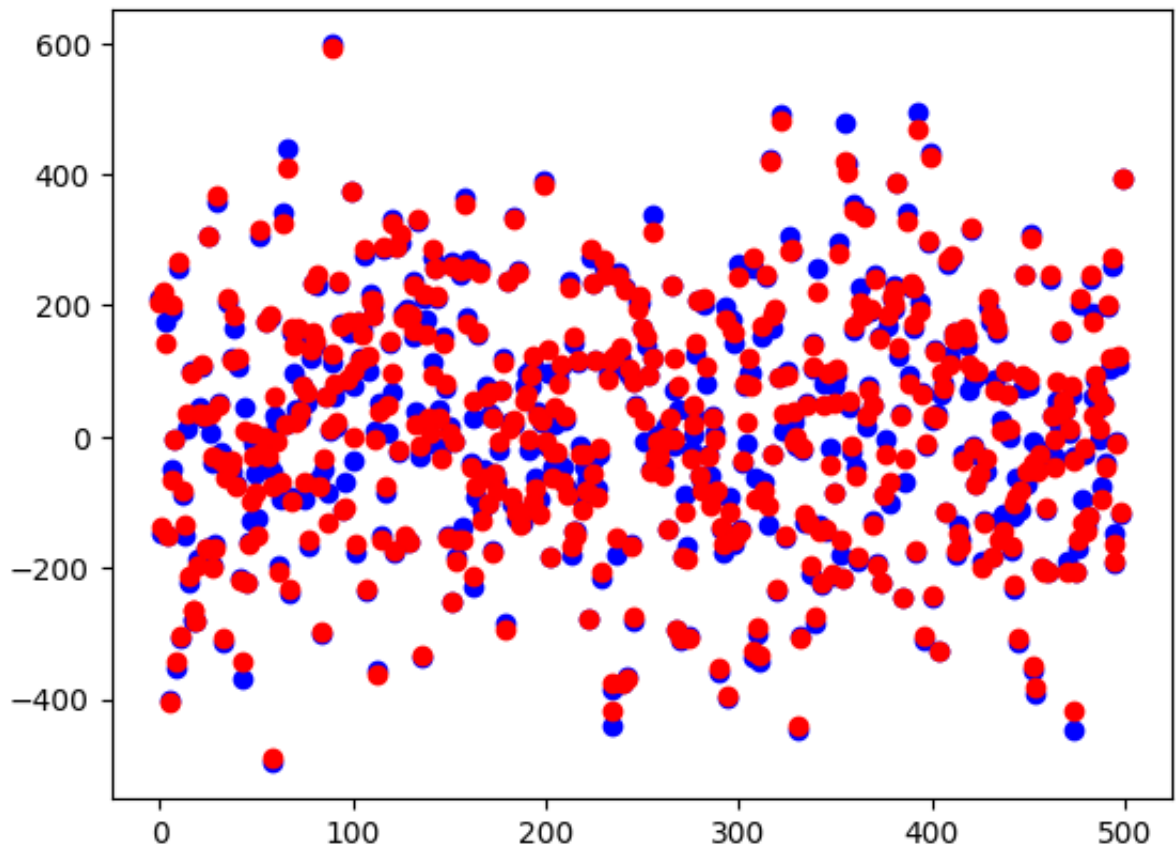
```

<matplotlib.collections.PathCollection at 0x7990f69cf260>



```
1 # Test
2 from matplotlib import pyplot as plt
3 plt.scatter(range(500),y_test,c='blue')
4 plt.scatter(range(500),pred,c='red')
```

<matplotlib.collections.PathCollection at 0x7990f6344440>



```
1 len(regr.coefs_)
```

3

```
1 regr.coefs_[0].shape
```

(100, 15)

```
1 regr.intercepts_[2].shape
```

(1,)

